

INTRODUCTION TO VERSION CONTROL

The Digital Time Machine — Grade 6 Session 1 Guide

Subject: Computer Science & Coding

Level: Grade 6

Provider: Planrs Academy

1. Welcome, Code Explorers!

Have you ever worked on a school project for hours, only to accidentally delete it? Or maybe you made a change you hated and couldn't go back? **Version Control** is here to save the day! Think of it as a professional "Undo" button for everything you create on a computer.



Explorer Tip: Always remember that in Git, nothing is ever truly lost. It's your ultimate safety net for coding!

2. The Problem: File Name Chaos

We've all seen files named `Project_Final.doc`, `Project_Final_v2.doc`, and `Project_REAL_FINAL_I_PROMISE.doc`. This is messy, confusing, and makes it almost impossible to know which version is actually the latest! Version control replaces this chaos with a clean, organized, and time-stamped history of your work.

3. What is Git vs. GitHub?

Git: Your Personal Digital Notebook

Git is a tool installed on your computer. It acts like a private diary for your code. Every time you finish a task, you tell Git to take a "snapshot" of your work. If you make a mistake tomorrow, you can jump back to the snapshot you took today.

GitHub: The National Library for Code

If Git is your private notebook, GitHub is like a massive public library. It is an online platform where you can "push" your work so teammates or people across the world can see, collaborate on, and improve your projects.

4. Essential Terms: Repository & Commit

- **The Repository (Repo):** Your Digital Almirah or school locker. It's a special folder where Git tracks every file, change, and snapshot safely backed up by date and time.
- **The Commit:** Saving with a purpose! A commit takes a snapshot of your project. Unlike a standard "Save", a commit requires a **Commit Message** explaining *what* changed and *why*.



Indian Context: Space Missions at ISRO

Scientists at ISRO use version control systems for satellite and rocket launch software. Imagine if they lost critical code mid-flight! By tracking every single line of code and revision, ISRO engineers ensure missions like Chandrayaan and Aditya-L1 are safe, accurate, and successful.

5. Writing Good Commit Messages

Bad commit messages like "stuff", "abc", or "fixed" are unhelpful for you and your team. Good commit messages tell a story!

-  **Bad:** "changed things"
-  **Good:** "Changed website background color to dark blue"

6. Branching: Parallel Worlds

Sometimes you want to test a brand new feature without breaking your working project. **Branching** lets you create a separate path—like a "choose your own adventure" book where you can explore one storyline without altering the original book!

Classroom Activity: "Human Git"

Pass a single sheet of paper around the class. Each student adds one sentence to a continuous story. Before passing it, write a commit message on a sticky note (e.g., "Added a space dragon to Chapter 1"). This mimics how developers track changes together!

WORKSHEET: VERSION CONTROL IN ACTION

Practice tracking, committing, and organizing your code like a professional developer!

Activity 1: File Name Chaos vs. Git Commits

Convert these chaotic file names into professional Commit Messages:

Messy File Name	Clear Commit Message
Science_Project_Final_v3_fixed.doc	<i>e.g., Added conclusion section and fixed formatting</i>
Game_Code_NEW_FINAL_REALLY.py	
Robot_Movement_v2_working_pls.cpp	

Activity 2: Commit History & Time Machine Tracker

You are building a school library website. Fill out your project's commit timeline:

Commit #	What Feature Did You Add/Fix?	Commit Message
Commit 1	Created basic home page header	"Initial commit: Created home page layout"
Commit 2		

Commit #	What Feature Did You Add/Fix?	Commit Message
Commit 3		

Activity 3: Branching Strategy ("Choose Your Own Adventure")

1. What is your main project topic?

2. Describe a experimental "Branch" feature you want to test without breaking your main project:

STUDENT PRACTICE ASSESSMENT

Review & Concept Mastery Questions — Grade 6 Version Control

Instructions: Answer all questions clearly in the spaces provided or on a separate answer sheet. No reference answers are provided on this sheet.

Section A: Multiple Choice Questions

Q1. What is the primary function of Git?

- [A] A social media app for video sharing
- [B] A local version control system that acts like a digital time machine for code
- [C] A hardware component inside a computer
- [D] A search engine for downloading games

Q2. What is the key difference between Git and GitHub?

- [A] Git is online; GitHub is only on your computer
- [B] Git tracks changes locally on your computer; GitHub is an online platform for sharing and collaboration
- [C] Git is used only by students; GitHub is used only by ISRO
- [D] There is no difference between them

Q3. What must be included whenever you make a "Commit" in Git?

- [A] A password
- [B] A commit message explaining what was changed
- [C] A payment
- [D] An image file

Q4. What does "Branching" allow developers to do?

- [A] Delete the entire repository permanently
- [B] Test new ideas on a separate path without affecting the main project
- [C] Turn off the internet connection
- [D] Rename all files to "Final.doc"

Section B: Short Answer & Concept Application

Q5. Why is saving files with names like `Project_Final_v2_REAL.doc` problematic, and how does version control solve this issue?

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Q6. Why is it critical for organizations like ISRO to use version control during satellite or space rocket missions?

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Q7. Explain why "Fixed button color on navigation bar" is a better commit message than "updated code".

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Section C: Scenario-Based Problem Solving

Q8. Scenario: Aarav is building a website for his school science fair. The home page works perfectly. Now he wants to add a complex interactive 3D map, but he is scared that adding the code might break his working website.

- (a) What Git feature should Aarav use before writing the new 3D map code?
- (b) If his 3D map code fails completely, how does version control protect his original working home page?

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